Part A – 1-D Arrays (Q1–Q6)
1. Write a Java program to input and display 5 integers using a 1-D array.

```
import java.util.Scanner;

public class ArrayInputDisplay {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[5];

        System.out.println("Enter 5 integers:");

        // Input elements
        for (int i = 0; i < 5; i++) {
            numbers[i] = sc.nextInt();
        }

        System.out.println("The entered integers are:");

        // Display elements
        for (int i = 0; i < 5; i++) {
            System.out.println(numbers[i]);
        }

        sc.close();
    }
}
```

OUTPUT:
Enter 5 integers:
10
20
30
40
50

The entered integers are:
10
20
30
40
50

2. Create a Java program to input 10 integers and display them in reverse order.

```
import java.util.Scanner;

public class ReverseArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[] numbers = new int[10];

        System.out.println("Enter 10 integers:");

        // Input elements
        for (int i = 0; i < 10; i++) {
            numbers[i] = sc.nextInt();
        }

        System.out.println("Numbers in reverse order:");

        // Display in reverse order
        for (int i = 9; i >= 0; i--) {
            System.out.print(numbers[i] + " ");
        }

        sc.close();
    }
}
```

OUTPUT:
Enter 10 integers:
10 20 30 40 50 60 70 80 90 100

Numbers in reverse order:
100 90 80 70 60 50 40 30 20 10

3. Write a Java program to find the sum of all elements in a 1-D array.

```
import java.util.Scanner;

public class SumArrayElements {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();

        int[] numbers = new int[n];
        int sum = 0;

        System.out.println("Enter " + n + " elements:");

        for (int i = 0; i < n; i++) {
            numbers[i] = sc.nextInt();
            sum = sum + numbers[i];
        }

        System.out.println("Sum of all elements = " + sum);

        sc.close();
    }
}
```

OUTPUT:
Enter the number of elements: 5
Enter 5 elements:
10 20 30 40 50
Sum of all elements = 150

4. Create a Java program to find the maximum element in an array.

```
import java.util.Scanner;

public class MaximumElement {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();

        int[] numbers = new int[n];

        System.out.println("Enter " + n + " elements:");

        for (int i = 0; i < n; i++) {
            numbers[i] = sc.nextInt();
        }
```

```
        int max = numbers[0];

        for (int i = 1; i < n; i++) {
            if (numbers[i] > max) {
                max = numbers[i];
            }
        }

        System.out.println("Maximum element = " + max);

        sc.close();
    }
}
```

OUTPUT:
Enter the number of elements: 5
Enter 5 elements:
10 45 20 80 35
Maximum element = 80

5. Write a Java program to count even and odd numbers in an array.

```
import java.util.Scanner;

public class EvenOddCount {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();

        int[] numbers = new int[n];

        int evenCount = 0;
        int oddCount = 0;

        System.out.println("Enter " + n + " integers:");

        for (int i = 0; i < n; i++) {
            numbers[i] = sc.nextInt();
```

```
            if (numbers[i] % 2 == 0) {
                evenCount++;
            } else {
                oddCount++;
            }
        }

        System.out.println("Number of even elements: " + evenCount);
        System.out.println("Number of odd elements: " + oddCount);

        sc.close();
    }
}
```

OUTPUT:
Enter the number of elements: 6
Enter 6 integers:
10 15 20 25 30 35
Number of even elements: 3
Number of odd elements: 3

6. Create a Java program to copy all elements from one array to another.

```
import java.util.Scanner;

public class CopyArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();

        int[] original = new int[n];
        int[] copy = new int[n];

        System.out.println("Enter " + n + " elements:");

        // Input elements into original array
        for (int i = 0; i < n; i++) {
            original[i] = sc.nextInt();
        }

        // Copy elements
```

```
        for (int i = 0; i < n; i++) {
            copy[i] = original[i];
        }

        System.out.println("Elements of copied array:");

        for (int i = 0; i < n; i++) {
            System.out.print(copy[i] + " ");
        }

        sc.close();
    }
}
```

OUTPUT:
Enter the number of elements: 5
Enter 5 elements:
10 20 30 40 50

Elements of copied array:
10 20 30 40 50

Part B – 2-D Arrays (Q7–Q13)
7. Write a Java program to input and display elements of a 3 × 3 matrix.

```
import java.util.Scanner;

public class MatrixInputDisplay {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][] matrix = new int[3][3];

        System.out.println("Enter 9 elements for the 3 x 3 matrix:");

        // Input matrix elements
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                matrix[i][j] = sc.nextInt();
            }
        }

        System.out.println("The 3 x 3 Matrix is:");
```

```
        // Display matrix elements
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(matrix[i][j] + " ");
            }
            System.out.println();
        }

        sc.close();
    }
}
```

OUTPUT:
Enter 9 elements for the 3 x 3 matrix:
1 2 3
4 5 6
7 8 9

The 3 x 3 Matrix is:
1 2 3
4 5 6
7 8 9

8. Create a Java program to find the sum of all elements in a matrix.

```
import java.util.Scanner;

public class MatrixSum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        System.out.print("Enter number of columns: ");
        int columns = sc.nextInt();

        int[][] matrix = new int[rows][columns];
        int sum = 0;

        System.out.println("Enter matrix elements:");
```

```
        // Input and calculate sum
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                matrix[i][j] = sc.nextInt();
                sum = sum + matrix[i][j];
            }
        }

        System.out.println("Sum of all matrix elements = " + sum);

        sc.close();
    }
}
```

OUTPUT:

```
Enter number of rows: 2
Enter number of columns: 3
Enter matrix elements:
1 2 3
4 5 6
Sum of all matrix elements = 21
```

9. Write a Java program to find the sum of each row in a matrix.

```
import java.util.Scanner;

public class RowSumMatrix {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        System.out.print("Enter number of columns: ");
        int columns = sc.nextInt();

        int[][] matrix = new int[rows][columns];

        System.out.println("Enter matrix elements:");

        // Input matrix elements
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
```

```
            matrix[i][j] = sc.nextInt();
        }
    }

    // Calculate sum of each row
    for (int i = 0; i < rows; i++) {
        int sum = 0;

        for (int j = 0; j < columns; j++) {
            sum = sum + matrix[i][j];
        }

        System.out.println("Sum of row " + (i + 1) + " = " + sum);
    }

    sc.close();
  }
}
```

OUTPUT:
Enter number of rows: 3
Enter number of columns: 3
Enter matrix elements:
1 2 3
4 5 6
7 8 9

Sum of row 1 = 6
Sum of row 2 = 15
Sum of row 3 = 24

10. Create a Java program to find the sum of each column in a matrix.

```
import java.util.Scanner;

public class ColumnSumMatrix {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter number of rows: ");
    int rows = sc.nextInt();

    System.out.print("Enter number of columns: ");
```

```
        int columns = sc.nextInt();

        int[][] matrix = new int[rows][columns];

        System.out.println("Enter matrix elements:");

        // Input matrix elements
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                matrix[i][j] = sc.nextInt();
            }
        }

        // Calculate sum of each column
        for (int j = 0; j < columns; j++) {
            int sum = 0;

            for (int i = 0; i < rows; i++) {
                sum = sum + matrix[i][j];
            }

            System.out.println("Sum of column " + (j + 1) + " = " + sum);
        }

        sc.close();
    }
}
```

OUTPUT:
Enter number of rows: 3
Enter number of columns: 3
Enter matrix elements:
1 2 3
4 5 6
7 8 9

Sum of column 1 = 12
Sum of column 2 = 15
Sum of column 3 = 18

11. Write a Java program to transpose a matrix.

```
import java.util.Scanner;
```

```
public class ColumnSumMatrix {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        System.out.print("Enter number of columns: ");
        int columns = sc.nextInt();

        int[][] matrix = new int[rows][columns];

        System.out.println("Enter matrix elements:");

        // Input matrix elements
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                matrix[i][j] = sc.nextInt();
            }
        }

        // Calculate sum of each column
        for (int j = 0; j < columns; j++) {
            int sum = 0;

            for (int i = 0; i < rows; i++) {
                sum = sum + matrix[i][j];
            }

            System.out.println("Sum of column " + (j + 1) + " = " + sum);
        }

        sc.close();
    }
}
```

OUTPUT:

```
Enter number of rows: 2
Enter number of columns: 3
Enter matrix elements:
1 2 3
4 5 6
```

Original Matrix:
1 2 3
4 5 6

Transpose Matrix:
1 4
2 5
3 6

12. Create a Java program to add two matrices of the same size.

```
import java.util.Scanner;

public class MatrixAddition {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        System.out.print("Enter number of columns: ");
        int columns = sc.nextInt();

        int[][] matrix1 = new int[rows][columns];
        int[][] matrix2 = new int[rows][columns];
        int[][] sum = new int[rows][columns];

        System.out.println("Enter elements of first matrix:");

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                matrix1[i][j] = sc.nextInt();
            }
        }

        System.out.println("Enter elements of second matrix:");

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                matrix2[i][j] = sc.nextInt();
            }
        }
```

```
        // Add matrices
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                sum[i][j] = matrix1[i][j] + matrix2[i][j];
            }
        }

        System.out.println("Sum of the matrices:");

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                System.out.print(sum[i][j] + " ");
            }
            System.out.println();
        }

        sc.close();
    }
}
```

OUTPUT:

```
Enter number of rows: 2
Enter number of columns: 2

Enter elements of first matrix:
1 2
3 4

Enter elements of second matrix:
5 6
7 8

Sum of the matrices:
6 8
10 12
```

13. Write a Java program to multiply two matrices (basic case with 3 × 3 matrices).

```
import java.util.Scanner;

public class MatrixMultiplication {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        int[][] matrix1 = new int[3][3];
        int[][] matrix2 = new int[3][3];
        int[][] result = new int[3][3];

        System.out.println("Enter elements of first 3 x 3 matrix:");

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                matrix1[i][j] = sc.nextInt();
            }
        }

        System.out.println("Enter elements of second 3 x 3 matrix:");

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                matrix2[i][j] = sc.nextInt();
            }
        }

        // Matrix multiplication
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                for (int k = 0; k < 3; k++) {
                    result[i][j] += matrix1[i][k] * matrix2[k][j];
                }
            }
        }

        System.out.println("Result of Matrix Multiplication:");

        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(result[i][j] + " ");
            }
            System.out.println();
        }

        sc.close();
    }
}
```

OUTPUT:
Enter elements of first 3 x 3 matrix:
1 2 3
4 5 6
7 8 9

Enter elements of second 3 x 3 matrix:
1 0 0
0 1 0
0 0 1

Result of Matrix Multiplication:
1 2 3
4 5 6
7 8 9

Part C – 3-D Arrays (Q14–Q18)
14.Write a Java program to input and display elements of a 2 × 2 × 2 3-D array.

```
import java.util.Scanner;

public class ThreeDArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][][] array = new int[2][2][2];

        System.out.println("Enter 8 elements for the 2 x 2 x 2 array:");

        // Input elements
        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                for (int k = 0; k < 2; k++) {
                    array[i][j][k] = sc.nextInt();
                }
            }
        }

        System.out.println("\nElements of the 3-D array:");

        // Display elements
        for (int i = 0; i < 2; i++) {
```

```
            for (int j = 0; j < 2; j++) {
                for (int k = 0; k < 2; k++) {
                    System.out.println(
                        "array[" + i + "][" + j + "][" + k + "] = "
                        + array[i][j][k]
                    );
                }
            }
        }

        sc.close();
    }
}
```

OUTPUT:
Enter 8 elements for the 2 x 2 x 2 array:
1 2 3 4 5 6 7 8

Elements of the 3-D array:
array[0][0][0] = 1
array[0][0][1] = 2
array[0][1][0] = 3
array[0][1][1] = 4
array[1][0][0] = 5
array[1][0][1] = 6
array[1][1][0] = 7
array[1][1][1] = 8

15. Create a Java program to find the sum of all elements in a 3-D array.

```
import java.util.Scanner;

public class Sum3DArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][][] array = new int[2][2][2];
        int sum = 0;

        System.out.println("Enter 8 elements for the 2 x 2 x 2 array:");

        // Input elements and calculate sum
        for (int i = 0; i < 2; i++) {
```

```
            for (int j = 0; j < 2; j++) {
                for (int k = 0; k < 2; k++) {
                    array[i][j][k] = sc.nextInt();
                    sum += array[i][j][k];
                }
            }
        }

        System.out.println("Sum of all elements = " + sum);

        sc.close();
    }
}
```

OUTPUT:
Enter 8 elements for the 2 x 2 x 2 array:
1 2 3 4 5 6 7 8

Sum of all elements = 36

16. Write a Java program to count positive and negative numbers in a 3-D array.

```
import java.util.Scanner;

public class PositiveNegative3D {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][][] array = new int[2][2][2];

        int positiveCount = 0;
        int negativeCount = 0;

        System.out.println("Enter 8 elements for the 2 x 2 x 2 array:");

        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                for (int k = 0; k < 2; k++) {
                    array[i][j][k] = sc.nextInt();

                    if (array[i][j][k] > 0) {
                        positiveCount++;
                    } else if (array[i][j][k] < 0) {
```

```
                    negativeCount++;
                }
            }
        }
    }

    System.out.println("Positive numbers: " + positiveCount);
    System.out.println("Negative numbers: " + negativeCount);

    sc.close();
  }
}
```

OUTPUT:
Enter 8 elements for the 2 x 2 x 2 array:
10 -5 20 -8 15 -3 0 25

Positive numbers: 4
Negative numbers: 3

17. Create a Java program to find the maximum element in a 3-D array.

```
import java.util.Scanner;

public class Max3DArray {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int[][][] array = new int[2][2][2];

      System.out.println("Enter 8 elements for the 2 x 2 x 2 array:");

      // Input elements
      for (int i = 0; i < 2; i++) {
         for (int j = 0; j < 2; j++) {
            for (int k = 0; k < 2; k++) {
               array[i][j][k] = sc.nextInt();
            }
         }
      }

      int max = array[0][0][0];
```

```
        // Find maximum element
        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                for (int k = 0; k < 2; k++) {
                    if (array[i][j][k] > max) {
                        max = array[i][j][k];
                    }
                }
            }
        }

        System.out.println("Maximum element = " + max);

        sc.close();
    }
}
```

OUTPUT:
Enter 8 elements for the 2 x 2 x 2 array:
10 25 8 45 12 30 5 20

Maximum element = 45

18. Write a Java program to search for an element in a 3-D array and display its position.

```
import java.util.Scanner;

public class Search3DArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int[][][] array = new int[2][2][2];

        System.out.println("Enter 8 elements for the 2 x 2 x 2 array:");

        // Input elements
        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                for (int k = 0; k < 2; k++) {
                    array[i][j][k] = sc.nextInt();
                }
            }
        }
```

```
        System.out.print("Enter element to search: ");
        int search = sc.nextInt();

        boolean found = false;

        // Search element
        for (int i = 0; i < 2; i++) {
            for (int j = 0; j < 2; j++) {
                for (int k = 0; k < 2; k++) {
                    if (array[i][j][k] == search) {
                        System.out.println("Element found at position: "
                                + "[" + i + "][" + j + "][" + k + "]");
                        found = true;
                    }
                }
            }
        }

        if (!found) {
            System.out.println("Element not found.");
        }

        sc.close();
    }
}
```

OUTPUT:
Enter 8 elements for the 2 x 2 x 2 array:
10 20 30 40 50 60 70 80

Enter element to search: 50
Element found at position: [1][0][0]

Part D – Jagged Arrays (Q19–Q20)
19. Write a Java program to create and display a jagged array with 3 rows of different lengths.

```
public class JaggedArray {
    public static void main(String[] args) {

        // Create jagged array with 3 rows of different lengths
        int[][] array = {
            {10, 20},
```

```
            {30, 40, 50, 60},
            {70, 80, 90}
        };

        System.out.println("Elements of Jagged Array:");

        // Display jagged array
        for (int i = 0; i < array.length; i++) {
            for (int j = 0; j < array[i].length; j++) {
                System.out.print(array[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

OUTPUT:
Elements of Jagged Array:
10 20
30 40 50 60
70 80 90

20. Create a Java program to store marks for students where each student has taken a different number of subjects, then display them.

```
import java.util.Scanner;

public class StudentMarks {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of students: ");
        int students = sc.nextInt();

        int[][] marks = new int[students][];

        // Input marks
        for (int i = 0; i < students; i++) {
            System.out.print("Enter number of subjects for student " + (i + 1) + ": ");
            int subjects = sc.nextInt();

            marks[i] = new int[subjects];
```

```
            System.out.println("Enter marks:");

            for (int j = 0; j < subjects; j++) {
                marks[i][j] = sc.nextInt();
            }
        }

        // Display marks
        System.out.println("\nStudent Marks:");

        for (int i = 0; i < students; i++) {
            System.out.print("Student " + (i + 1) + ": ");

            for (int j = 0; j < marks[i].length; j++) {
                System.out.print(marks[i][j] + " ");
            }

            System.out.println();
        }

        sc.close();
    }
}
```

OUTPUT:

```
Enter number of students: 3

Enter number of subjects for student 1: 3
Enter marks:
80 75 90

Enter number of subjects for student 2: 2
Enter marks:
85 70

Enter number of subjects for student 3: 4
Enter marks:
60 75 80 90

Student Marks:
Student 1: 80 75 90
Student 2: 85 70
Student 3: 60 75 80 90
```

Part E - Additional Questions
21. Write a program to rotate an array to the left by one position.

```
import java.util.Scanner;

public class LeftRotateArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();

        int[] array = new int[n];

        System.out.println("Enter array elements:");

        for (int i = 0; i < n; i++) {
            array[i] = sc.nextInt();
        }

        // Store first element
        int first = array[0];

        // Shift all elements one position to the left
        for (int i = 0; i < n - 1; i++) {
            array[i] = array[i + 1];
        }

        // Place first element at the last position
        array[n - 1] = first;

        System.out.println("Array after left rotation:");

        for (int i = 0; i < n; i++) {
            System.out.print(array[i] + " ");
        }

        sc.close();
    }
}
```

OUTPUT:
Enter number of elements: 5

Enter array elements:
10 20 30 40 50

Array after left rotation:
20 30 40 50 10

22. Count how many times each element appears in a 1-D array.

```
import java.util.Scanner;

public class ElementFrequency {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();

        int[] array = new int[n];
        boolean[] visited = new boolean[n];

        System.out.println("Enter array elements:");

        for (int i = 0; i < n; i++) {
            array[i] = sc.nextInt();
        }

        System.out.println("Frequency of each element:");

        for (int i = 0; i < n; i++) {

            if (visited[i]) {
                continue;
            }

            int count = 1;

            for (int j = i + 1; j < n; j++) {
                if (array[i] == array[j]) {
                    count++;
                    visited[j] = true;
                }
            }
```

```
            System.out.println(array[i] + " appears " + count + " time(s).");
        }

        sc.close();
    }
}
```

OUTPUT:
Enter number of elements: 7
Enter array elements:
10 20 10 30 20 10 40

Frequency of each element:
10 appears 3 time(s).
20 appears 2 time(s).
30 appears 1 time(s).
40 appears 1 time(s).

23. Find the second largest element in a 1-D array.

```
import java.util.Scanner;

public class SecondLargest {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();

        int[] array = new int[n];

        System.out.println("Enter array elements:");

        for (int i = 0; i < n; i++) {
            array[i] = sc.nextInt();
        }

        int largest = Integer.MIN_VALUE;
        int secondLargest = Integer.MIN_VALUE;

        for (int i = 0; i < n; i++) {
            if (array[i] > largest) {
                secondLargest = largest;
```

```
            largest = array[i];
        } else if (array[i] > secondLargest && array[i] != largest) {
            secondLargest = array[i];
        }
    }

    if (secondLargest == Integer.MIN_VALUE) {
        System.out.println("Second largest element does not exist.");
    } else {
        System.out.println("Second largest element = " + secondLargest);
    }

    sc.close();
  }
}
```

OUTPUT:
Enter number of elements: 5
Enter array elements:
10 45 20 80 35
Second largest element = 45

24. Remove duplicate elements from a 1-D array (without using collections).

```
import java.util.Scanner;

public class RemoveDuplicates {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter number of elements: ");
    int n = sc.nextInt();

    int[] array = new int[n];

    System.out.println("Enter array elements:");

    for (int i = 0; i < n; i++) {
        array[i] = sc.nextInt();
    }

    int[] unique = new int[n];
    int count = 0;
```

```
        // Find unique elements
        for (int i = 0; i < n; i++) {
            boolean isDuplicate = false;

            for (int j = 0; j < count; j++) {
                if (array[i] == unique[j]) {
                    isDuplicate = true;
                    break;
                }
            }

            if (!isDuplicate) {
                unique[count] = array[i];
                count++;
            }
        }

        System.out.println("Array after removing duplicates:");

        for (int i = 0; i < count; i++) {
            System.out.print(unique[i] + " ");
        }

        sc.close();
    }
}
```

OUTPUT:
Enter number of elements: 7
Enter array elements:
10 20 10 30 20 40 30

Array after removing duplicates:
10 20 30 40

25. Find the sum of the main diagonal and secondary diagonal elements of a square matrix.

```
import java.util.Scanner;

public class DiagonalSum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter size of square matrix: ");
        int n = sc.nextInt();

        int[][] matrix = new int[n][n];

        System.out.println("Enter matrix elements:");

        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                matrix[i][j] = sc.nextInt();
            }
        }

        int mainDiagonalSum = 0;
        int secondaryDiagonalSum = 0;

        for (int i = 0; i < n; i++) {
            mainDiagonalSum += matrix[i][i];
            secondaryDiagonalSum += matrix[i][n - 1 - i];
        }

        System.out.println("Sum of main diagonal = " + mainDiagonalSum);
        System.out.println("Sum of secondary diagonal = " + secondaryDiagonalSum);

        sc.close();
    }
}
```

OUTPUT:
Enter size of square matrix: 3
Enter matrix elements:
1 2 3
4 5 6
7 8 9

Sum of main diagonal = 15
Sum of secondary diagonal = 15

26. Display the upper and lower triangular parts of a square matrix separately.

```
import java.util.Scanner;
```

```
public class TriangularMatrix {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter size of square matrix: ");
        int n = sc.nextInt();

        int[][] matrix = new int[n][n];

        System.out.println("Enter matrix elements:");

        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                matrix[i][j] = sc.nextInt();
            }
        }

        // Upper triangular matrix
        System.out.println("\nUpper Triangular Matrix:");

        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i <= j) {
                    System.out.print(matrix[i][j] + " ");
                } else {
                    System.out.print("0 ");
                }
            }
            System.out.println();
        }

        // Lower triangular matrix
        System.out.println("\nLower Triangular Matrix:");

        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i >= j) {
                    System.out.print(matrix[i][j] + " ");
                } else {
                    System.out.print("0 ");
                }
            }
            System.out.println();
        }
```

```
        sc.close();
    }
}
```

OUTPUT:
Enter size of square matrix: 3
Enter matrix elements:
1 2 3
4 5 6
7 8 9

Upper Triangular Matrix:
1 2 3
0 5 6
0 0 9

Lower Triangular Matrix:
1 0 0
4 5 0
7 8 9

27. Print only the boundary elements of a matrix.

```
import java.util.Scanner;

public class BoundaryElements {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        System.out.print("Enter number of columns: ");
        int columns = sc.nextInt();

        int[][] matrix = new int[rows][columns];

        System.out.println("Enter matrix elements:");

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                matrix[i][j] = sc.nextInt();
```

```
            }
        }

        System.out.println("Boundary Elements:");

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {

                if (i == 0 || i == rows - 1 ||
                    j == 0 || j == columns - 1) {

                    System.out.print(matrix[i][j] + " ");
                } else {
                    System.out.print("  ");
                }
            }
            System.out.println();
        }

        sc.close();
    }
}
```

OUTPUT:
Enter number of rows: 4
Enter number of columns: 4

Enter matrix elements:
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16

Boundary Elements:
1 2 3 4
5 8
9 12
13 14 15 16

28. Search for an element in a 2-D array and display its row & column position.

```
import java.util.Scanner;
```

```
public class Search2DArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        System.out.print("Enter number of columns: ");
        int columns = sc.nextInt();

        int[][] matrix = new int[rows][columns];

        System.out.println("Enter matrix elements:");

        // Input matrix elements
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                matrix[i][j] = sc.nextInt();
            }
        }

        System.out.print("Enter element to search: ");
        int search = sc.nextInt();

        boolean found = false;

        // Search element
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                if (matrix[i][j] == search) {
                    System.out.println("Element found at:");
                    System.out.println("Row = " + (i + 1));
                    System.out.println("Column = " + (j + 1));
                    found = true;
                }
            }
        }

        if (!found) {
            System.out.println("Element not found.");
        }

        sc.close();
    }
```

```
}
```

OUTPUT:
Enter number of rows: 3
Enter number of columns: 3

Enter matrix elements:
10 20 30
40 50 60
70 80 90

Enter element to search: 50

Element found at:
Row = 2
Column = 2

29. Sort the elements of each row in a jagged array.

```
import java.util.Scanner;

public class SortJaggedArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        int[][] array = new int[rows][];

        // Input jagged array
        for (int i = 0; i < rows; i++) {
            System.out.print("Enter number of elements in row " + (i + 1) + ": ");
            int size = sc.nextInt();

            array[i] = new int[size];

            System.out.println("Enter elements for row " + (i + 1) + ":");

            for (int j = 0; j < size; j++) {
                array[i][j] = sc.nextInt();
            }
        }
```

```
        // Sort each row using Bubble Sort
        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < array[i].length - 1; j++) {
                for (int k = 0; k < array[i].length - 1 - j; k++) {
                    if (array[i][k] > array[i][k + 1]) {
                        int temp = array[i][k];
                        array[i][k] = array[i][k + 1];
                        array[i][k + 1] = temp;
                    }
                }
            }
        }

        // Display sorted jagged array
        System.out.println("\nSorted Jagged Array:");

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < array[i].length; j++) {
                System.out.print(array[i][j] + " ");
            }
            System.out.println();
        }

        sc.close();
    }
}
```

OUTPUT:

```
Enter number of rows: 3

Enter number of elements in row 1: 3
Enter elements for row 1:
30 10 20

Enter number of elements in row 2: 4
Enter elements for row 2:
50 20 40 10

Enter number of elements in row 3: 2
Enter elements for row 3:
90 60

Sorted Jagged Array:
```

10 20 30
10 20 40 50
60 90

30. Check if a given matrix is a sparse matrix (more zero elements than non-zero).

```
import java.util.Scanner;

public class SparseMatrix {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of rows: ");
        int rows = sc.nextInt();

        System.out.print("Enter number of columns: ");
        int columns = sc.nextInt();

        int[][] matrix = new int[rows][columns];
        int zeroCount = 0;
        int nonZeroCount = 0;

        System.out.println("Enter matrix elements:");

        for (int i = 0; i < rows; i++) {
            for (int j = 0; j < columns; j++) {
                matrix[i][j] = sc.nextInt();

                if (matrix[i][j] == 0) {
                    zeroCount++;
                } else {
                    nonZeroCount++;
                }
            }
        }

        System.out.println("Number of zero elements: " + zeroCount);
        System.out.println("Number of non-zero elements: " + nonZeroCount);

        if (zeroCount > nonZeroCount) {
            System.out.println("The matrix is a Sparse Matrix.");
        } else {
```

```
            System.out.println("The matrix is not a Sparse Matrix.");
        }

        sc.close();
    }
}
```

OUTPUT:

```
Enter number of rows: 3
Enter number of columns: 3

Enter matrix elements:
0 0 3
0 4 0
0 0 5

Number of zero elements: 6
Number of non-zero elements: 3
The matrix is a Sparse Matrix.
```